

January 2025 | 8.50 pm | B-216

Statistical Network Models

Session 1 – Welcome and Introduction



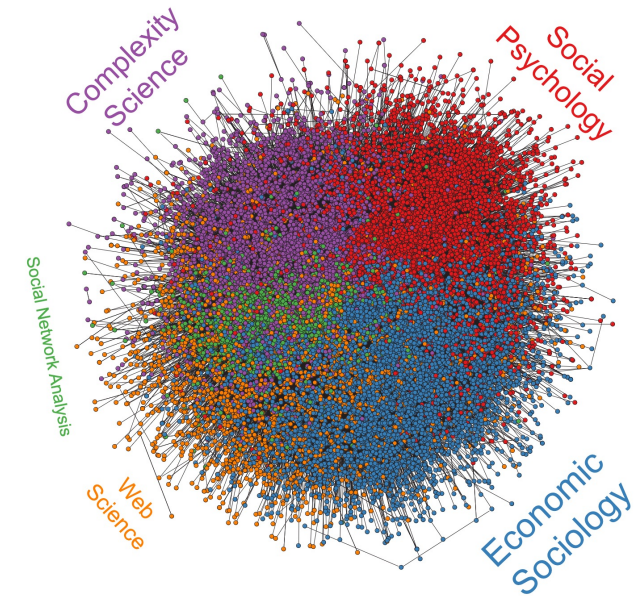
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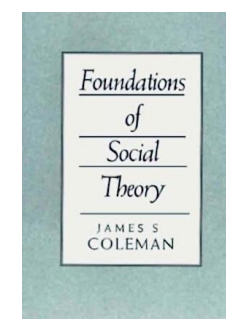
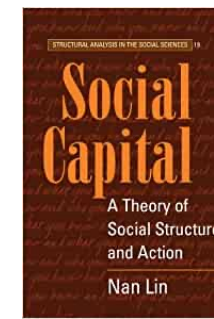
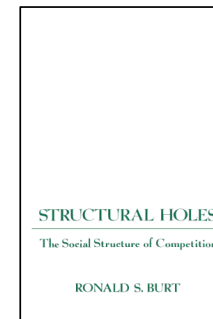
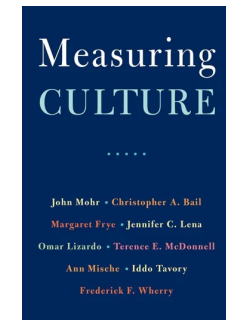
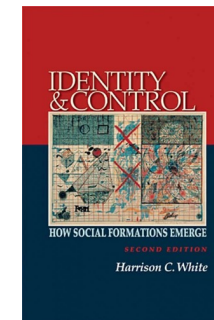
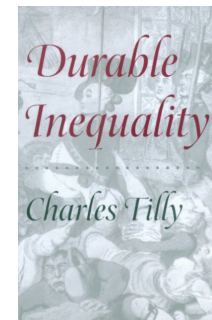
Session 1 - Introduction

- Overview of the course and organizational information
- Some thoughts on the differences between network science and social network analysis
- Theory and social networks
- Outlook next session



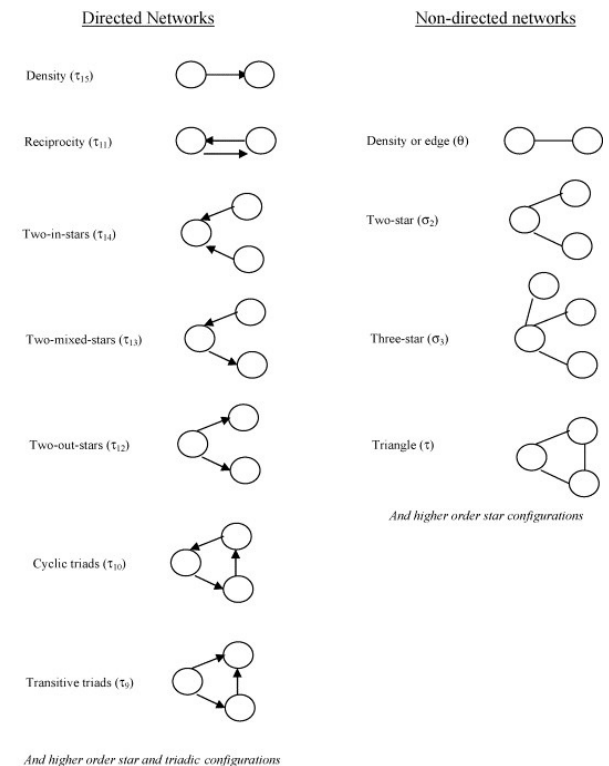
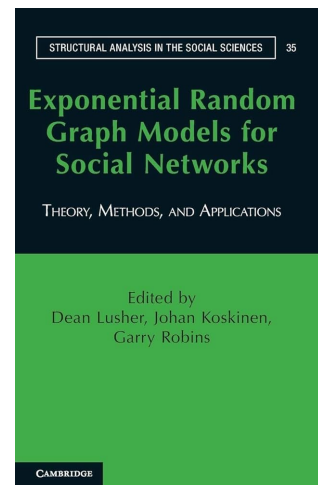
Session 2 – Sociological Network Theories

- Different strands of sociological theory
- Formalist vs. relational accounts
- Multi-paradigmatic perspective



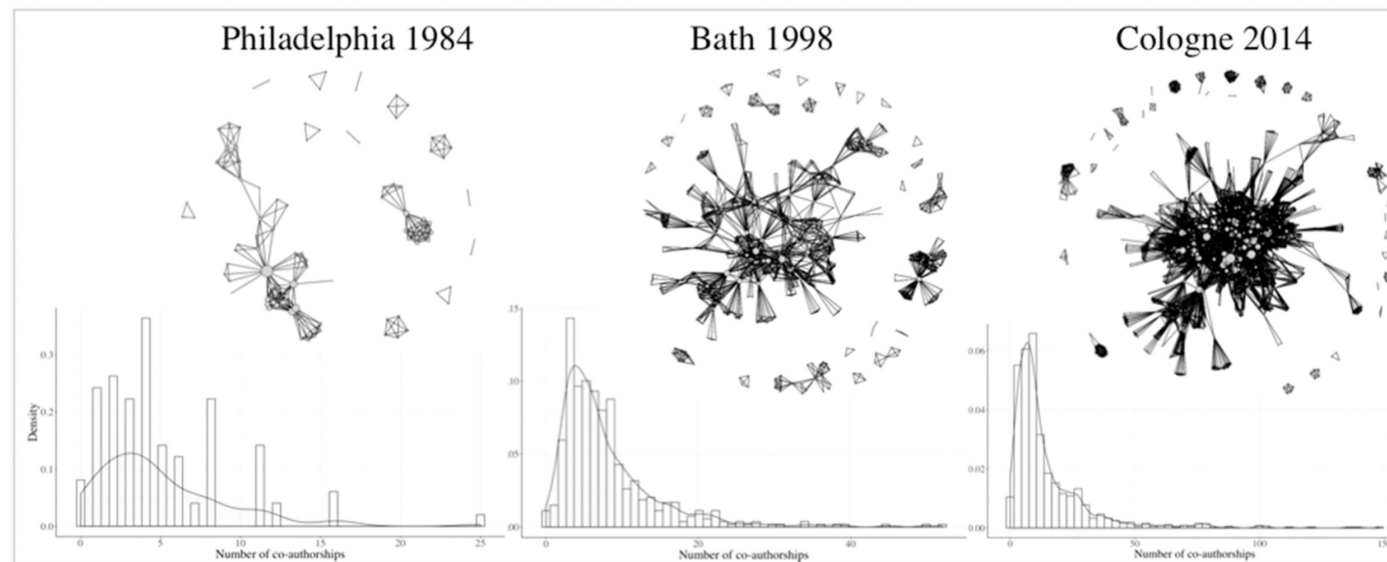
Session 3 – Statistical Network Models I

- Model for cross-sectional data, exponential random graph models (ERGMs)
- Methodological background of the models, discussion of link function, estimation procedures etc.



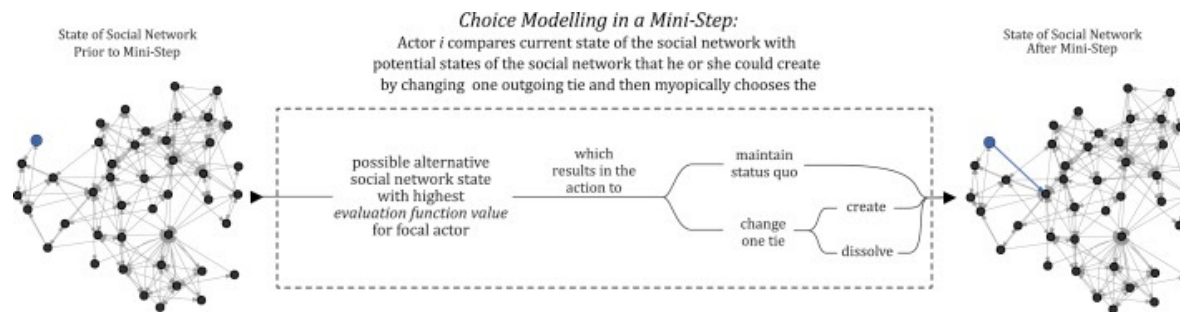
Session 4-/5 – Example Applications

- Using ERGMs to map structural changes in scientific and artistic collaboration networks over time



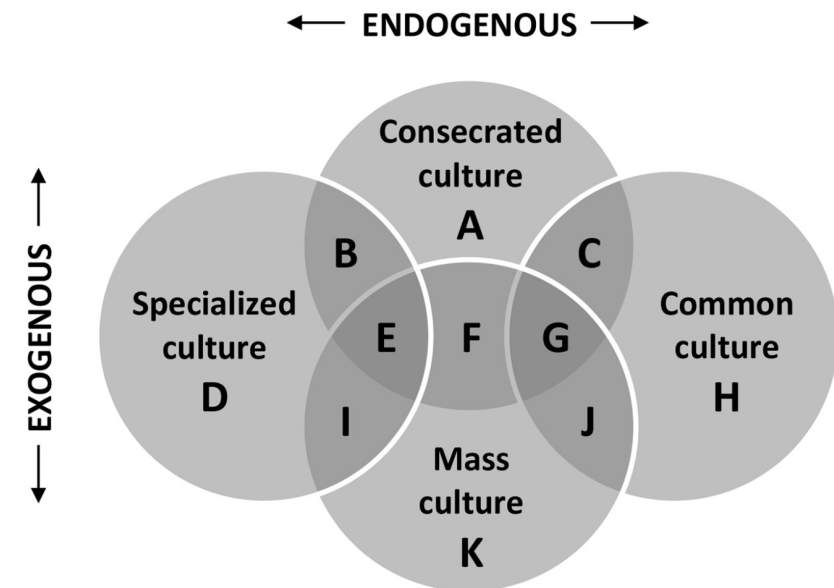
Session 6 – Statistical Network Models II

- Introduction to stochastic actor-oriented models (SAOMs) for longitudinal network analysis
- Methodological background of the models, discussion of link function, estimation procedures etc.



Session 7/8 –Application of SAOMs

- Reading a study that applies SAOMs to Facebook networks and cultural preferences
- I hope to present an example from a cultural field (e.g., collaboration network among jazz musicians)
- Fallback option: student networks and cultural preferences, depending on feasibility



Sessions 9 – Relational Hyper Event Models (RHEMs)

- Workshop session with Jürgen Lerner
- Examples from our research

Sessions 10 – Cognitive Science meets Network Analysis

- Bonus session with our TA Lotta Pesonen
- Showcasing the intersection of network analysis, cognitive science, and neuroscience

Sessions 11/12 – Student presentations

- Presentation by students, ~15-20 min

Tasks and grading

- Attendance of seminar and active participation (15%), pass/ fail
- Presentation (20%); ~15-20min, graded
- Research report (65%); approximately 3,500-5,000 words excluding tables and figures, graded
- Students may form groups of up to three students
- **Deadline for written report is April 7**

Potential data sources

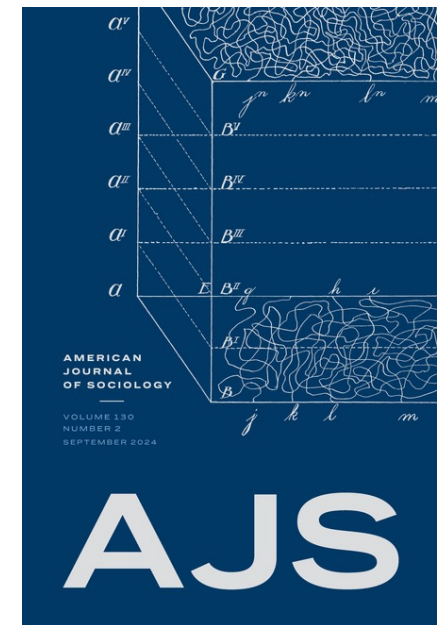
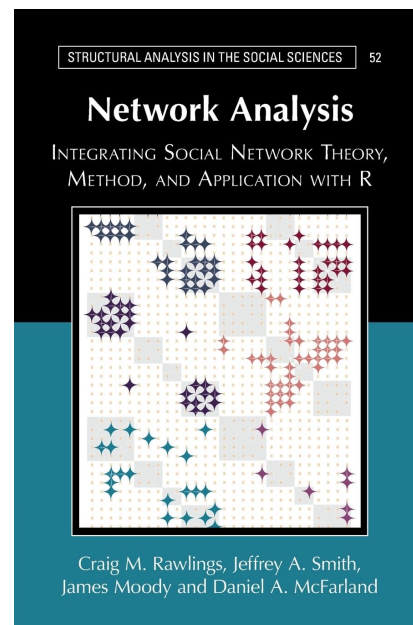
Use your **own data**, or get something from i.e.:

- SocioPatterns datasets: <http://www.sociopatterns.org/datasets/>
- Stanford Large Network Dataset Collection: <http://snap.stanford.edu/data/>
- The Colorado Index of Complex Networks: <https://icon.colorado.edu/#!/>
- Data Science Central: <https://www.datasciencecentral.com/profiles/blogs/great-github-list-of-public-data-sets>
- Kdnuggets: <https://www.kdnuggets.com/datasets/index.html>

SocioPatterns

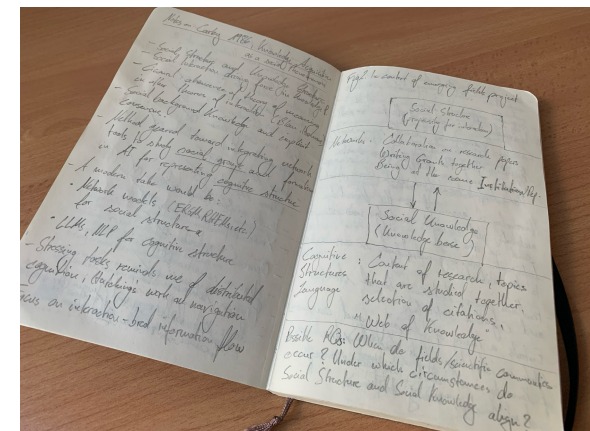


Key readings for this class



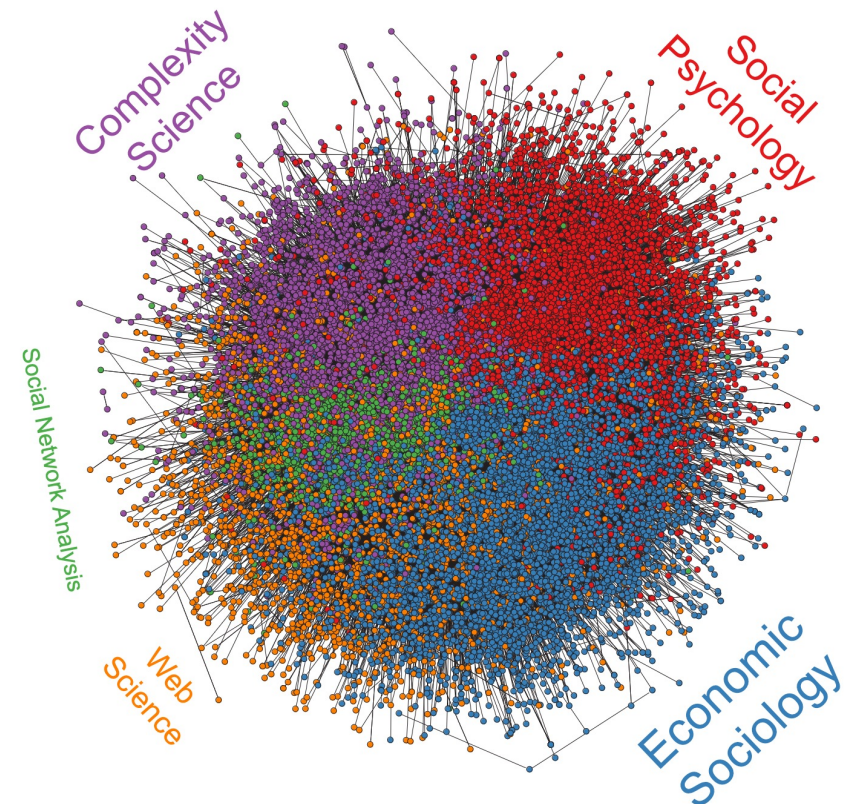
Reading and writing advice

- Cultivate reading and writing as a habit, find a place and time for it each day, avoid distractions, try to read on paper not digitally, print and buy books used or borrow them in the library, take notes, don't read word-by-word
- Write down the overall structure and key points of the text after you read it, write down any thoughts / associations on other material that you encountered during your study, write by hand, don't use a laptop or tablet, talk to others about your readings



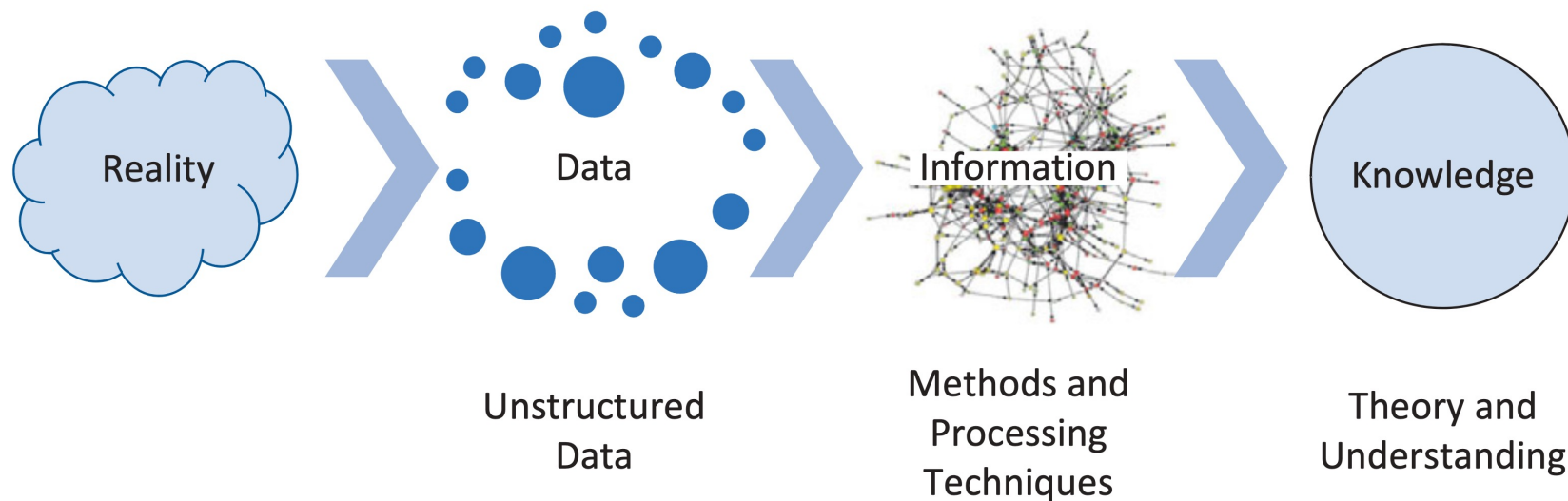
A note on the history of network science

- Harvard school (Harrison White, Block models) and California school (Freeman, centrality measures) in the 1970s
- Cultural turn in sociological network analysis (Breiger) and blossoming of structuralist approaches (e.g., Burt) in 1980-1990s
- From 2000s onward: physicists and computer scientists start network science



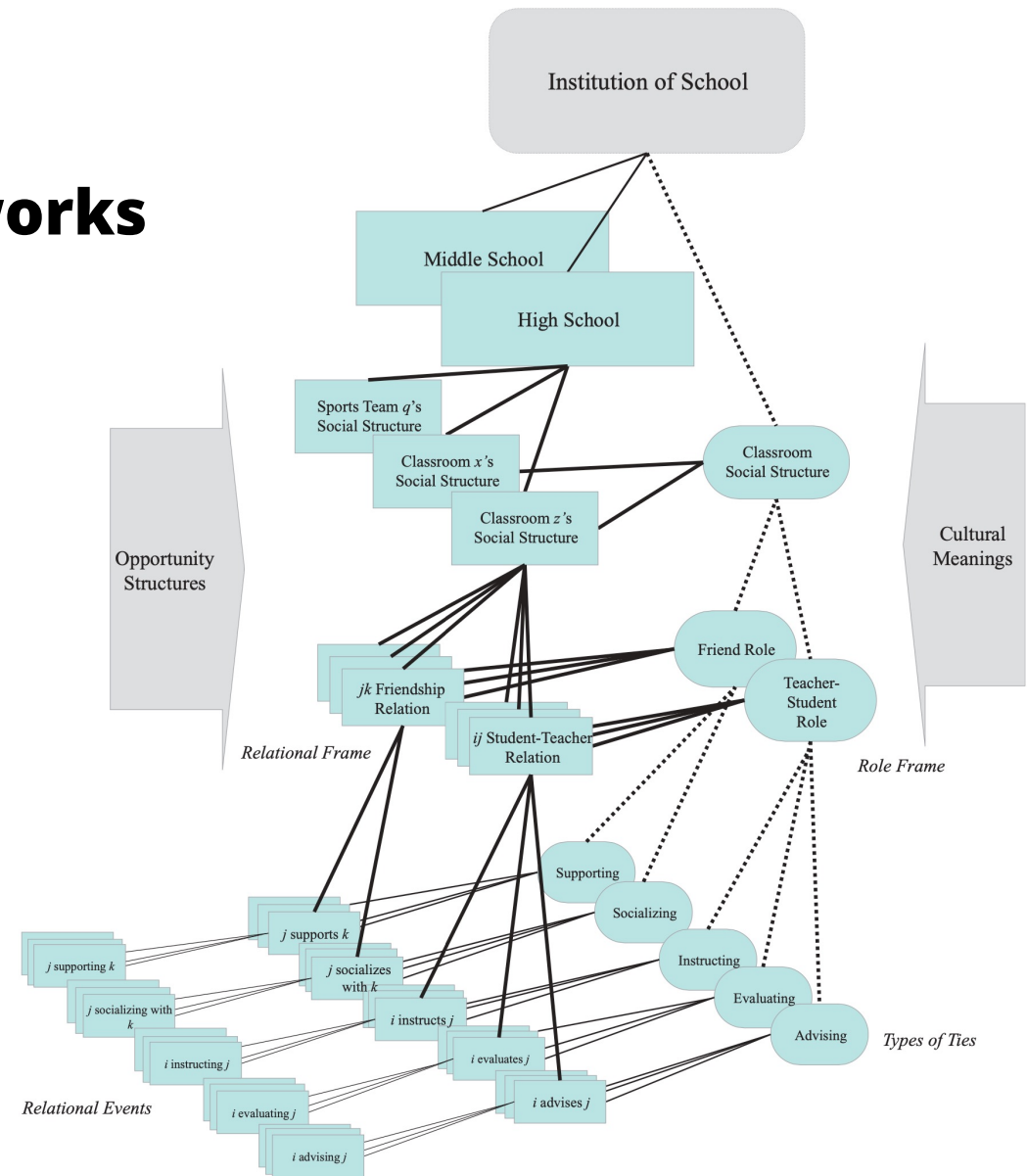
Network analysis in the social sciences

- What is the role of theory in social network analysis according to Rawlings et al. (2023)



Abstraction and social networks

- What is the figure on the right expressing?
- How is this approach to think about social networks different from what you encounter in network science and complex systems literature?

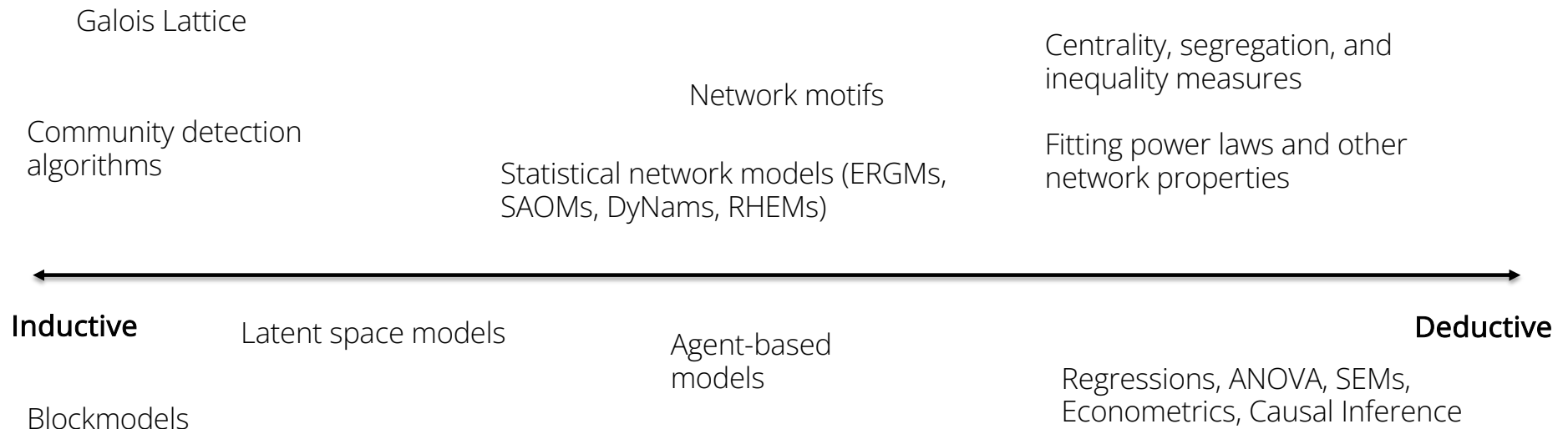


Research agendas in network research

- What are the differences between the research agendas outlines in the table below?
- Can you think of examples for the different boxes?

Perspective	<u>Networks as cause</u>	<u>Networks as consequence</u>
<u>Connectionist</u> <i>Networks as pipes</i>	Diffusion Peer influence Social capital	Social integration Peer selection Segregation
<u>Positional</u> <i>Networks as roles</i>	Popularity effects Role behavior Network constraint	Exchange pattern Network stability Career paths

Different network methods



Readings for next week

Fuhse, J. A. (2020). Theories of Social Networks. ***The Oxford Handbook of Social Networks***, 34-49.

Erikson, E. (2013). Formalist and relationalist theory in social network analysis. ***Sociological theory***, **31**(3), 219-242.

Thank you for your attention!